

Module-3. (Propositional Logic)

⇒ What is Discrete Mathematics?

→ Part of mathematics devoted to the study of discrete objects. (No exact definition). But we know what comes under it and what not.

Integers, People, etc. comes under discrete objects & Real Nos, Temp, etc. are continuous objects and are studied in Calculus Subject.

Discrete Mathematics is the backbone of Comp. Science. Also known as Mathematics for Comp. Science.

⇒ Mathematical Logic: Also called language for mathematics.

It gives precise, unambiguous meanings to mathematical statement or theorems, etc.

→ Reason we don't use natural languages like English for computers and mathematics is that these languages are ambiguous.

For Ex: ~~In stat~~ S1: if ab is even then a is even or b is even.
S2: attack city a or city b .

In S1 we can have both a is even & b is even also true, but in S2, this is not the case, there we allow exactly one. So, in mathematical logic we will use " $\vee$ " in S1 and " $\oplus$ " in S2 and there will be no ambiguity.

Ex:2 In statement, "The sun is shining and I feel happy" we are not clear if both clauses are independent or other clause is due to first. So again ambiguity.

So, mathematical logic resolves the ambiguity of Natural languages.

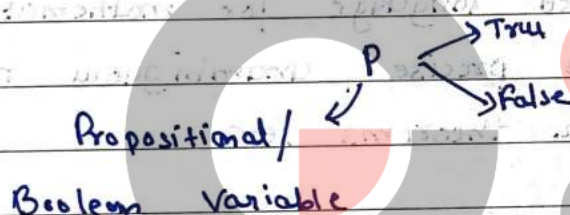
* Various Types of Mathematical Logic:

- Propositional Logic, (Gate Syllabus)
- First Order Logic, ("...") / Predicate Logic / Predicate-Calculus
- Second Order Logic,
- Third order Logic,
- fuzzy logic, etc

* Propositional Logic → Simplest Mathematical Logic.

→ World of True / False.

↳ Every var/exp. will be True or False



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⇒ Proposition: A proposition is a declarative sentence (i.e. a sentence that declares a fact) that can be either true or false; it must be one or the other, and it cannot be both. It can also be called statement.

It is a declarative sentence to which it is possible to assign a value of either true or false.

Prop.	Truth Val.
Ex: → "Jaipur is Capital of India"	False
→ "All cows are brown"	False
→ $2 \times 2 = 5$	False
→ "Earth is further from the sun than Venus"	True

↳ Not req. to know that if it is a prop or not

→ There are 1000 stones on Mars True/False only

Prop $\rightarrow$ Proposition
 Pv $\rightarrow$ Proposition Variable.

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Assertive

If a sentence can be either True or False then it is proposition. THAT'S ALL.

Some sentence that are NOT propositions:

- $\rightarrow$ Commands (Ex: Fly, you fools!)
- $\rightarrow$ Questions (Ex: What are you doing?)
- $\rightarrow$ $x+2x=-2$ is also NOT proposition bcz. x is a free var.

Que: Which of them is a proposition?

a) $4n = 2 + 3n$

b) for all integers n , $2n$ is an Even Number.

$\Rightarrow$ a) is not bcz. we can't say it is True or false w/o knowing value of n . (free variable)

b) It is always true. (n is variable but it is not free).

$\Rightarrow$ S: "S is false" is not a proposition, it is a paradox.
 If we consider S true then "S is false" is True and S is false and vice versa. It is neither true nor false.
 So it is a paradox. And for prop. either true or false is required.

* Propositional Variable: Each proposition will be represented by a propositional variable.

Generally repr. as lower case letters such as p, q, r, s, etc.
 Each var. can take either true or false (Any one).

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★ Atomic Proposition: It is a prop. whose truth or falsity does not depend on the truth or falsity of any other Prop.

Ex: p : 4 is a prime number (Truth Value $\rightarrow$ False)
 prop. var. q : New Delhi is Capital of India (Truth Value $\rightarrow$ True).

Now At

"4 is a prime no, and, ND is capital of India" is a prop. but not atomic prop. as Truth val of this depends on other prop.

→ NOTE Every atomic prop. can be represented using single prop. variable, since it does not reduce further to other more basic prop.

→ New prop. can be created from atomic prop. with the help of logical connectives (operators).

Ex: p AND q , p OR q , $p \oplus q$, etc.

★ Compound Prop: Prop. that are formed from existing prop. using logical ^{operators} are called compound prop.

Ex: 4 is a prime no, & ND is capital of India, etc.

⇒ Standard Logical operators / Connectives: (8)

→ NOT (only unary op)

→ BI-Implication

→ AND

→ NAND

→ OR

→ NOR

→ Exclusive OR

→ Implication

(We will see these all in detail)

Start syll. will be covered in this

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→ If we know the truth value of atomic prop., then we can get truth value of compound propositions. Atomic prop.'s truth value should be given in question.

→ Negation is only std. conn. that is unary, rest all are binary i.e. they operate on two propositions.

I LC1- Negation/Not ($\neg$): Let p be a prop. The negation of p is "It is not the case that p ".

Ex: P : This book is cool.
 $\neg P \equiv \bar{P} \equiv P' \equiv \neg P$: This book is not cool. / @

Most
Std. Symbol

It is not the case that this book is cool.

It is not true that this book is cool.

It is false that this book is cool.

This book is cool, is false.

→ The truth value of $\neg P$ is the ^{opp. of} truth value of P .

P	$\neg P$
T	F
F	T

II LC2- Conjunction/AND ($\wedge$): Let p & q be prop. The conjunction of p and q , denoted by $P \wedge Q$, is the prop. "p and q". $P \wedge Q$ is true when both p and q are true and false otherwise. ($P \wedge P, P \wedge Q, P \wedge \text{false}$)

Ex: P : This book is cool.

Q : I am at home.

$P \wedge Q$: This book is cool and I am at home.

Truth Table: (TT) It tells us about the truth value of a combination of truth values of atomic prop.

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TT →

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

Both of them should be True.

⇒ English words which translates to Conjunction: Use intuition ^{BTW}

Even Though, ^(comma) And, However, Yet, though, Moreover, Nevertheless, But, etc. [Use your own intuition]

III Disjunction (OR ($\vee$)): Let p & q be propositions. The disjunction of p and q , denoted by $p \vee q$, is the proposition " p or q ". It is false when both p & q are false and is true otherwise. ($P \vee Q$, $P \vee Q$)

Ex: p : This book is interesting.

q : I am staying at home.

$P \vee q$: This book is interesting or I am staying at home.

Truth Table:

P	Q	$P \vee Q$
T	F	T
F	T	T
T	T	T
F	F	F

Atleast one should be True.

Also called inclusive OR because it involves T/T as also True. In ex-OR T/T is not True.

"You can eat or play" is Inc. OR.

"You can eat or play but Not both" is Exclusive OR.

So for OR & XOR we can see English is ambiguous but Prop. logic is unambiguous.

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IV. Exclusive OR (XOR) ($\oplus$): Let p & q be prop. The exclusive or of p and q , denoted by $p \oplus q$, is the proposition that is true when exactly one of p and q is true and false otherwise.

Ex: p : You can come on Sunday

q : You can come on Monday

$p \oplus q$: You can come on Sunday or Monday but not both.

Truth Table

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

Exactly one of them should be True.

⇒ Also can be written as either p or q for exclusive or. in some books but we will consider it as inclusive or.

V. NAND / Negation of AND ($\uparrow$): $p \uparrow q = \neg(p \wedge q)$

Truth Table:

p	q	$p \uparrow q$
T	T	F
T	F	T
F	T	T
F	F	T

Atleast one should be false.

VI. NOR / Negation of OR ($\downarrow$): $p \downarrow q = \neg(p \vee q)$

Truth Table:

p	q	$p \downarrow q$
T	T	F
T	F	F
F	T	F
F	F	T

Both of them should be false.

*** Imp

VII Implication / Conditional Statements ($\rightarrow$): Let p and q be propositions.

The conditional statement

$p \rightarrow q$ is the proposition "if p then q ". The conditional statement $p \rightarrow q$ is false when p is true and q is false, and true otherwise. Here p is called the hypothesis (or antecedent, or premise) and q is called the conclusion (or consequence).

Statement $p \rightarrow q$ is called a conditional statement because $p \rightarrow q$ asserts that q is true on the condition that p holds.

Ex: P : Today is Sunday. Q : I am Happy. $P \rightarrow Q$: If today is Sunday, then I am happy.If P then $Q \equiv P$ implies $Q \equiv P \rightarrow Q$.Imp.
Truth Table.

P	Q	$P \rightarrow Q$
F	T	T
F	F	T
T	T	T
T	F	F

It will be false only if P is true & Q is false. If P is false any Q will work.

- - - - - only false condition.

→ Assume an Airline makes the Claim:

"If you fly with us, then you will get free food".

This claim will be FALSE iff you fly and you don't get food i.e. P is true and Q is false, else the claim will be True always.

★ $P \rightarrow Q$ means if "P is true then Q is true as well". It says nothing about what happens if P is false. And above statement will be false if we know P is true & Q is false.

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Ex:2 Political party's claim. "If we win, then you will get job". will be false only if they ~~don't~~ win and I didn't get job.

→ Implication is like a promise or claim. That will be only false in If P then Q. if P is True and Q is false.

$\downarrow$ Hypothesis $\downarrow$ Conclusion

→ Conditional Statements are One way bcz. if P is true then Q is true is meaning of $P \rightarrow Q$, but if Q is true then we don't know about P, that's why one way statement it is.

P = True

P = True

P = False

 $\rightarrow$ $\leftarrow$ $\rightarrow$

Q = True

Q = True

Q = False

One way.

Ex: If you are in Delhi, you are in India. is obviously one way.

→ $P \rightarrow Q \equiv Q \leftarrow P$ both means "P implies Q".

→ $P \wedge Q \equiv Q \wedge P$, $P \vee Q \equiv Q \vee P$, $P \oplus Q \equiv Q \oplus P$, $P \rightarrow Q \neq Q \rightarrow P$.

ie.

P	Q	$P \rightarrow Q$	$Q \rightarrow P$
T	T	T	T
T	F	F	T
F	T	F	F
F	F	T	T

Ex: P: You are PM of India.

Q: You are Indian.

$P \rightarrow Q$ is True here i.e. if P then Q.

$Q \rightarrow P$ is False here i.e. if Q then P.

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⇒ Important Points about Implication Operator:

→ $P \rightarrow Q$ is only false when P is True & Q is false, else always True.

→ $P \rightarrow Q$ is True when either P is ~~True~~ False or Q is ~~False~~ True.

⇒ $P \rightarrow Q \equiv$ If P true then Q True or
 $P \rightarrow Q \equiv$ If Q false then P false. $\equiv \neg Q \rightarrow \neg P$.

⇒ Necessary and Sufficient Conditions

→ P is sufficient for Q means, Happening of P is enough for happening of Q. If P happens then Q happens. i.e. If P is true then Q is True always and that simply means $P \rightarrow Q$.

It also means if Q did not happen, that means P also did not happen. $\neg Q \rightarrow \neg P$

→ Definition of Sufficient Condⁿ: A condition A is said to be sufficient for B, if and only if the truth of A guarantees the truth of B.

→ Ex:1 Being prime is sufficient for Being odd for Integers ≥ 2 .

Ex:2 Getting top 10 Rank is sufficient for IITB.

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→ P is necessary for Q means, without P, Q cannot happen. Not happening of P implies Not happening of Q.

$$\neg P \rightarrow \neg Q \Rightarrow Q \rightarrow P$$

→ So without P, Q cannot happen.

If P does not happen then surely Q doesn't happen.

If Q happens that means P happens. ($Q \rightarrow P$)

→ Ex:1 Fuel is necessary to drive car. (but it is not sufficient)

Ex:2 For int ≥ 2 ; Being odd is necessary for Being prime.

Ex:3 Air is necessary for being alive.

bez $\neg \text{Air} \rightarrow \neg \text{Life}$.

So $\text{Life} \rightarrow \text{Air}$.

→ α is suff. for β $\alpha \rightarrow \beta$

→ $\leftarrow$ is necessary for β $\beta \rightarrow \alpha$

→ α is suff. for $\beta \equiv \beta$ is necessary for α .

Ex: Being father is suff. for being male,

Being male is not suff. for being father.

Being ~~father~~ male is necessary for being father.

Being father is not necessary for being male.

→ Trick is " $\rightarrow$ " for suff and " $\leftarrow$ " for necessary.

→ Some ex which have neither suff. nor necessary.

Ex: Studying chem. is neither suff. nor necessary for Gate Exam.

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→ n being even is necessary as well as sufficient for nt_2 being even.

→ Definition of necessary condition: A condition A is said to be necessary for B if the falsity of A guarantees the falsity of B .

→ If α is sufficient for β , then β is necessary for α and vice versa. (Father, Male Example)

⇒ Various English translations of $P \rightarrow Q$.

→ if P then Q

→ P implies Q

→ when P, Q

→ Q if P

→ P is sufficient for Q

→ ~~sufficient~~ necessary condition for P is Q

→ Q is necessary for P

→ P only when Q

↳ feeling of necessity

→ If P, Q

→ Whenever P, Q

→ provided that P, Q

→ Q provided that P

→ Sufficient for Q is P

→ Q whenever P

→ Q follows from P

→ P only if Q

→ Q if P

→ which of the following is a prop? is a dumb que & this is waste of time. No good exam will ask this que.

→ $R: p \wedge q$, $p \vee q$ are prop variable & R is prop. formulae.

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★
VIII

Bisimplification / Biconditional Statements:

→ As we know that, Implication is used to repr. only one way statements.

Ex: P: You appear GATE

Q: You top GATE

Here $Q \rightarrow P$ ✓, but $P \rightarrow Q$ ✗ is false.

→ But sometimes statements are two ways.

Ex: P: Value is 2.

Q: Value is Even prime.

Here $P \rightarrow Q$ ✓ & $Q \rightarrow P$ ✓ also.

→ So, To repr. two ways statements, we have a new logical connective "Bi-Implication" ($\leftrightarrow$).

If $A \rightarrow B$ and $B \rightarrow A$, then $A \leftrightarrow B$.

So

$$A \leftrightarrow B \equiv (A \rightarrow B) \wedge (B \rightarrow A)$$

→ The biconditional connective $p \leftrightarrow q$ is read "p if and only if (also called iff) q".

→ Intuitively, either both p & q are true else none are true.

→ We can also think of it as equality i.e. both prop. p & q must have same truth value.

→ Truth Table.

P	q	$P \rightarrow q$	$q \rightarrow P$	$(P \rightarrow q) \wedge (q \rightarrow P)$ $P \leftrightarrow q$
F	F	T	T	T
F	T	T	F	F
T	F	F	T	F
T	T	T	T	T

True if both P & q are same

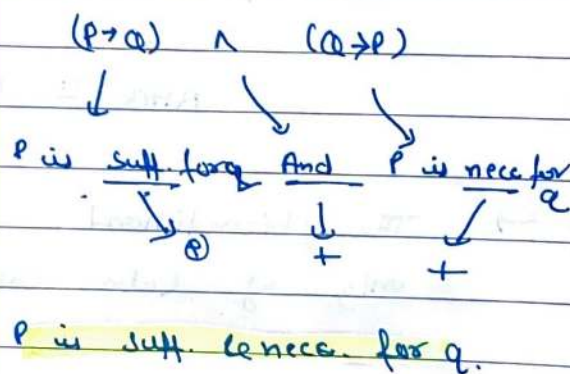
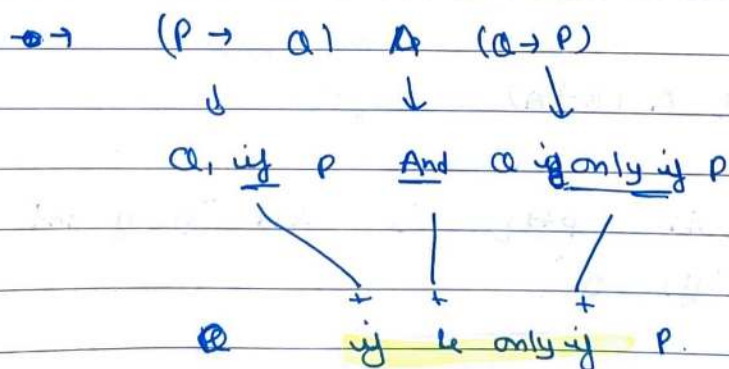
→ One thing that can be noticed from the above truth table is that $P \leftrightarrow q$ is negation of $P \oplus q$.

So, $P \leftrightarrow q = \neg(P \oplus q)$

or $P \oplus q = \neg(P \leftrightarrow q)$

→ Various English Translations for $P \leftrightarrow q$.

- P bimplication q .
- P implies q & q implies P .
- P implies q & vice versa.
- P iff q .
- P is suff. & nec. for P .
- P iff q , etc.

→ Also $P \leftrightarrow q$ is same as $q \leftrightarrow P$.

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★ Definition v/s Properties:

→ Definition is always if and only if (i.e. iff) statements, even if 'if' is written.

→ Implication tells us about property.

i.e. $P \rightarrow Q$ means Q is the property of P . / P has property Q .

Ex: $P =$ Prime no > 2 .

$Q =$ Being Odd

Here, $P \rightarrow Q$

So we can't define Prime no > 2 as odd numbers nor, we can say Prime no > 2 is property of Being odd.

We can only say that being odd is the property of prime number > 2 .

→ Bi-Implication tells us about definition.

i.e. $P \leftrightarrow Q$ means P is def. of Q & Q is def. of P .

Ex: $P =$ No. is 2.

$Q =$ No. is Even Prime.

Here $P \leftrightarrow Q$

So we can say being 2 is definition of being prime and also being prime is def. of being 2.

→ So $P \rightarrow Q$ says that P satisfy the property Q & $P \leftrightarrow Q$ says that only P satisfies the property Q .

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→ As we already know what is prop., we also know each prop. var. represent prop. atomic prop. Compound prop. can't be represented using prop. var.

→ Prop. formulae (Expression): These represent compound prop. also. These are formed with logical connectives & variables. Ex. if p & q are prop. var. then, $p \rightarrow q$, $p \wedge q$, etc.

→ Every prop. var. is also prop. formulae.

→ Every T, F are also "GO".

→ $p \wedge q$, $p \rightarrow q$, etc.

→ $p \wedge q$, $q \rightarrow$, $p \leftrightarrow$, $p \vdash$, etc. are not prop. formulae.

→ ~~Prop.~~ Prop. Logic is a world of True or False. It is a set/collection of all prop. formulae.

Prop. Logic $\left\{ \begin{array}{l} p, q, r, \dots \\ T, F \\ p \vee q, p \wedge q, p \rightarrow q, (p \rightarrow q) \rightarrow q, \text{ etc.} \end{array} \right.$

→ Prop. Formulae is same as compound prop. & prop. var. is same as atomic prop.

we found truth table of prop. formulae by taking all possible comb. of values of prop. var. present in it.

→ Prop. formulae is also called WFF (well framed formulae).

* Precedence Order of logical connectives:

$() > \neg > \wedge > \vee > \rightarrow > \leftrightarrow$
 $\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 Bracket Not Conjunction Disjunction Implication Bi-Implication

Ex: $p \wedge q \vee r \rightarrow (p \wedge q) \vee r$

→ Precedence order of $\oplus$, $\uparrow$ & $\downarrow$ (XOR, NAND and NOR) is not defined universally. Question will have proper parenthesis for these operators.

Ques: Let p repr. Truth statement, while q & r repr. False statement. Find truth value of following compound stmts.

a) $\sim [\sim p \wedge \sim q] \vee \sim q$

→ q is False $\Rightarrow \sim q$ is True $\Rightarrow$ True $\vee$ anything is True.

b) $\sim (p \wedge q) \wedge (r \vee \sim q)$

→ False

⇒ Truth Table: It tells us about the Truth value of a compound prop. for each combination of truth values of atomic prop.

Ex: 1

p	q	$p \wedge (p \rightarrow q)$
F	F	F
F	T	F
T	F	F
T	T	T

In $P \rightarrow Q$, $\neg P \vee Q = \text{True}$.

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Ex: 2 T.T. for $P \leftrightarrow (Q \vee R)$

P	Q	R	$Q \vee R$	$P \leftrightarrow (Q \vee R)$
F	F	F	F	T
F	F	T	T	F
F	T	F	T	F
F	T	T	T	F
T	F	F	F	F
T	F	T	T	T
T	T	F	T	T
T	T	T	T	T

→ Creating Truth tables is inefficient and time consuming method. We will rarely use it to solve questions. We will use analytical ways to solve questions quickly.

*** Three classes:

→ Tautology: It is a wff for which all Truth values are True. Ex: $P \vee \neg P$

→ Contradiction: It is a wff for which all Truth values are false. Ex: $P \wedge \neg P$

→ Contingency: It is a wff that is neither a tautology nor a contradiction i.e. at least one Truth value is True and at least one is false. Ex: $P \rightarrow Q$

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Que: Classify each of the below as tautology, contradiction or contingency, - If P is prop. var and T, F are constants

- | | | | |
|----------------------------|-------------|----------------------|---------------|
| a) P | Contingency | b) F | Contradiction |
| b) $\bar{P}$ | Contingency | i) $P \rightarrow F$ | Contingency |
| c) $T \rightarrow P$ | Contingency | | |
| d) $P \rightarrow P$ | Tautology | | |
| e) $P \rightarrow \bar{P}$ | Contingency | | |
| f) $\bar{P} \rightarrow P$ | Contingency | | |
| g) T | Tautology | | |

* $\rightarrow$ Given prop. formulae p , then p is a:

- tautology iff p always evaluates to true, regardless of how its variables are assigned.
- Contradiction iff p always evaluates to false, regardless of how its variables are assigned.
- Contingency iff p 's variable may be assigned in a way that makes p evaluate to true & may be assigned in another way that makes p evaluate to false.

$\rightarrow$ At least 1 True

$\rightarrow$ Satisfiable: if atleast truth value of 1 comb. is True.
(Tautology + Contingency). Also called Consistent.

$\rightarrow$ No True

$\rightarrow$ Unsatisfiable: no comb. can make True. i.e. Always False.
(Contradiction). Also called Fallacy/Inconsistent.

$\rightarrow$ At least 1 False

$\rightarrow$ Falsifiable: if at least truth val. of 1 comb. is False.
(Contradiction + Contingency). Also called Invalid.

$\rightarrow$ No False

$\rightarrow$ Non falsifiable: if no comb. can make it false.
(tautology). Also called valid.

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Summary:

All True / No False	Atleast one False / One True	All False / No True
tautology	Contingency	contradiction / fallacy
logically True	Both	logically False
Valid		Invalid
Satisfiable / consistent		Un Satisfiable / inconsistent
Unfalsifiable		Falsifiable
Satisfiable / Consistent		unsatisfiable / inconsistent

→ Solving questions by truth table is lengthy method, we will divide each formulae in two cases by any prop. var. i.e. if $p = \text{True}$ & $p = \text{False}$ and then derive final class of that formulae. This method is by Case method.

Ex: Let $\alpha = p \wedge \neg(p \vee q)$

Case I: $p = \text{True}$

$$T \wedge \neg(T \vee q)$$

$$T \wedge \neg(T)$$

$$T \wedge F$$

$$\underline{\underline{F}}$$

Case II: $p = F$

$$F \wedge \text{Anything}$$

$$\underline{\underline{F}}$$

So, it is contradiction.

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Ques: Prove following are all tautologies.

a) $(p \vee q) \leftrightarrow (q \vee p)$

$\Rightarrow$ Case I: $p = T$

$(T \vee q) \leftrightarrow (q \vee T)$

$T \leftrightarrow T$

T

Case II: $p = F$

$(F \vee q) \leftrightarrow (q \vee F)$

$q \leftrightarrow q$

TSo, both True $\Rightarrow$ Tautology

b) $q \rightarrow (p \vee q)$

$\Rightarrow$ Case I: $q = F$

$F \rightarrow (p \vee F)$

$F \rightarrow \text{anything}$

T

Case II: $q = T$

$T \rightarrow (p \vee T)$

$T \rightarrow T$

TSo, both True $\Rightarrow$ Tautology

c) $(p \wedge q) \rightarrow p$

$\Rightarrow$ Case I: $p = T$

$T \wedge q \rightarrow T$

$\text{anything} \rightarrow T$

T

Case II: $p = F$

$F \wedge q \rightarrow F$

$F \rightarrow F$

TSo, both true $\Rightarrow$ TautologyQues: Which of the following are tautology, contradiction & neither (i.e. contingency).

a) $\neg q \vee (p \wedge q)$

$\Rightarrow$ Case I: $q = T$

$F \vee (p \wedge T)$

$F \vee p$ ($p \wedge T = p$)

p

$(F \vee p = p)$

as p can be T or F , this is contingency

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b). $(\neg p \vee q) \vee (p \wedge q)$

$\Rightarrow$ Case I: $q = F$

$$(\neg p \vee F) \vee (p \wedge F)$$

$$\neg p \vee F$$

$\neg p$ as $\neg p$ can be T or F. It is contingency.

Que: Which of the following are Satisfiable (i.e. Not Contradiction)

a). $(p \wedge q) \wedge (\neg p \vee r) \wedge (\neg q \vee \neg r)$

$\Rightarrow$ Case I: $p = T$

$$(T \wedge q) \wedge (F \vee r) \wedge (\neg q \vee \neg r)$$

$$q \wedge r \wedge (\neg q \vee \neg r)$$

Case I.a: $p = T, q = T$

$$T \wedge r \wedge (F \vee \neg r)$$

$$r \wedge \neg r$$

$$\underline{\underline{F}}$$

Case I.b: $p = T, q = F$

$$F \wedge r \wedge (T \vee \neg r)$$

$$F \wedge T$$

$$\underline{\underline{F}}$$

$\Rightarrow$ Case II: $p = F$

$$(F \wedge q) \wedge (T \vee r) \wedge (\neg q \vee \neg r)$$

$$F \wedge T \wedge (\neg q \vee \neg r)$$

$$\underline{\underline{F}}$$

So as it is everytime false, it is not Satisfiable

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b) $(p \rightarrow q) \wedge (r \rightarrow p) \wedge (p \rightarrow r) \wedge (r \rightarrow p)$

$\Rightarrow$ Case I: $r = T$

$$(p \rightarrow T) \wedge (T \rightarrow p)$$

$$T \wedge p$$

$$\underline{p}$$

So if $p = T$, it will be True, So Satisfiable.

$$R = T$$

NOTE All of these que. can be also solved using Truth Table, but this is more analytical method. Use any in Exam.

★ Logical Equivalence: This is generally denoted by $\equiv$.

Two prop. α & β are said to be eq. i.e. $\alpha \equiv \beta$, iff α , β have equal truth value in every case. i.e. Both have same Truth table.

$\forall_1$	$\forall_2$	$\forall_3$	$\forall_4$	$\forall_5$	α	β
F	f	F	F	F	T	T
1					T	T
1					F	F
1					F	F
T	T	T	T	T	F	F

$\rightarrow$ This can also be seen as $\alpha \leftrightarrow \beta$ is always True. So, α & β are eq. if $\alpha \leftrightarrow \beta$ is Tautology.

$\rightarrow$ This is also same as $\alpha \equiv \beta$ iff $\alpha \leftrightarrow \beta$ is tautology and $\beta \rightarrow \alpha$ is Tautology.

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Ex: p : John is Intelligent q : John is Handworking $\neg(p \wedge q)$: It is not the case that John is both Intelligent & Handworking. $\neg p \vee \neg q$: John is not handworking or Not Intelligent!

How

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

p	q	$\neg(p \wedge q)$	$\neg p \vee \neg q$
F	F	T	T
F	T	T	T
T	F	T	T
T	T	F	F

→ The comp. prop p and q are called logically eq. if $p \leftrightarrow q$ is a tautology.

→ Some authors use ' $\Leftrightarrow$ ', instead of ' $\equiv$ ' for equivalence.

Ques:

$$\neg(p \rightarrow q) \equiv p \wedge (\neg q)$$

→ we can solve using Truth Table or by case method.

p	q	$\neg(p \rightarrow q)$	$p \wedge (\neg q)$
F	F	F	F
F	T	F	F
T	F	T	T
T	T	F	F

So. Both are eq.

$\alpha \equiv \beta$ if in every case $\alpha = \beta$

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By Case Method: Let $\alpha = \neg(p \rightarrow q)$
 $\beta = p \wedge \neg q$

Case I: $p = T$

$$\alpha = \neg q \quad (T \rightarrow q = q)$$

$$\beta = \neg q \quad (T \wedge q = q)$$

Case II: $p = F$

$$\alpha = \neg F \quad (F \rightarrow \text{anything} = T)$$

$$\beta = F \quad (F \wedge \text{anything} = F)$$

So $\alpha \equiv \beta$

Ques: Prove following compound prop. are logically eq.

a) $p \leftrightarrow (p \vee q)$ and $\neg p \vee q$

$\Rightarrow$ Let $\alpha = p \leftrightarrow (p \vee q)$ & $\beta = \neg p \vee q$

Case I: $p = \text{True}$

$$\alpha = q$$

$$\beta = q$$

Case II: $p = \text{false}$

$$\alpha = \text{True}$$

$$\beta = \text{True}$$

b) $\neg(p \vee (q \wedge (\neg p \rightarrow \neg r)))$ And $\neg p \wedge (q \rightarrow r)$
 α β

$\Rightarrow$ Case I: $p = T$

$$\alpha = F$$

$$\beta = F$$

Case II: $p = F$

$$\alpha = \neg(q \wedge r)$$

$$\beta = (q \rightarrow r)$$

Now we need to prove $\neg(q \wedge r) \equiv q \rightarrow r$
 but we already proved this above.

So $\alpha \equiv \beta$

$$\alpha \equiv \beta, \beta \equiv \gamma \Rightarrow \alpha \equiv \gamma$$

Ques: Prove that following expressions are logically equivalent:

$$\frac{p \rightarrow (q \vee r)}{\alpha}, \frac{(p \wedge (\neg q)) \rightarrow r}{\beta}, \frac{(p \wedge (\neg r)) \rightarrow q}{\gamma}$$

If we prove $\alpha \equiv \beta$ & $\alpha \equiv \gamma$, then $\beta \equiv \gamma$ is also

for $\alpha \equiv \beta$ there.

Case I: $P = T$

$$\alpha = q \vee r$$

$$\beta = \neg q \rightarrow r$$

Case Ia: $q = T$

$$\alpha = T$$

$$\beta = T$$

Case II: $P = F$

$$\alpha = F$$

$$\beta = F$$

So $\alpha \equiv \beta$

Case Ib: $q = F$

$$\alpha = r$$

$$\beta = r$$

Now for $\beta \equiv \gamma$

Case I, $P = T$

$$\alpha = q \vee r$$

$$\beta = \neg r \rightarrow q$$

And as we proved

above that $q \vee r \equiv \neg q \rightarrow r$

So, $r \vee q \equiv \neg r \rightarrow q$

So $\alpha \equiv \beta \equiv \gamma$

Case II: $P = F$

$$\alpha = F$$

$$\beta = F$$

So $\alpha \equiv \beta$

* Converse, Inverse and Contrapositive of Conditional Statements

- The prop. $q \rightarrow p$ is called the converse of $p \rightarrow q$.
- The prop. $\neg p \rightarrow \neg q$ is called the inverse of $p \rightarrow q$.
- The prop. $\neg q \rightarrow \neg p$ is called the contrapositive of $p \rightarrow q$.

(Remember it as we did proof by contraposition as $p \rightarrow q \equiv \neg q \rightarrow \neg p$)

Ex: p : Weather is nice.

q : I will wash car.

$p \rightarrow q$: If weather is nice, then I'll wash car.

(Converse) $q \rightarrow p$: If I will wash car, then weather will be nice.

(Inverse) $\neg p \rightarrow \neg q$: If weather is not nice, then I will not wash my car.

(Contrapositive) $\neg q \rightarrow \neg p$: If I will not wash car, then weather is not nice.

Contrapositive and Original Implication are equivalent and
Converse and Inverse of Original Implication are equivalent.

p	q	original $p \rightarrow q$	inverse $\neg p \rightarrow \neg q$	converse $q \rightarrow p$	Contrapositive $\neg q \rightarrow \neg p$
F	F	T	T	T	T
F	T	T	F	F	T
T	F	F	T	T	F
T	T	T	T	T	T

→ We can write formula of prop by checking their True Values.

Ex: $p \rightarrow q$ is True : p is False or q is True

$$\neg p \vee q \equiv \bar{p} + q$$

So

$$p \rightarrow q \equiv \bar{p} + q$$

Ex: 2 $P \oplus Q$ is True : $P=T, Q=F$ OR $Q=T, P=F$
 $= P\bar{Q} + Q\bar{P}$

Ex: 3 $P \leftrightarrow Q$ is True : $P=T, Q=T$ OR $P=F, Q=F$
 $= PQ + \bar{P}\bar{Q}$

Ex: 4 $\bar{P} \oplus Q$ is True : $\bar{P}=T, Q=T$ OR $\bar{P}=F, Q=F$
 $= P\bar{Q} + \bar{P}Q$

So $P \leftrightarrow Q = \bar{P} \oplus Q = P\bar{Q} + \bar{P}Q$

So, by this we can make formulas of any prop.

★ Unless word to Mathematical logic:

$P \text{ unless } Q \equiv P \text{ if not } Q \equiv \text{if not } Q, \text{ then } P$

So $P \text{ unless } Q = \neg Q \rightarrow P$

Ex: Jen won't go to the party unless Marry goes to the party
 is eq. to

If Marry doesn't go to the party, then Jen won't go to the party

Here it is not guaranteed that if Marry goes then Jen will go. He may not go even if Marry goes. But it is guaranteed that if Marry doesn't go, then Jen will surely not go.

Ex: 4 I will eat cake unless I'm full.
 is eq. to.

If I'm not full, then I will eat cake.

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So unless is eq. to If not.

Ex: I will not take if I am not full.

And also

provided that is eq. to If.

So $P \text{ unless } Q \equiv P \text{ if not } Q.$

$P \text{ provided that } Q \equiv P \text{ if } Q.$

Also W.K.T. $\neg A \rightarrow P \equiv P \vee A.$

So $\text{Unless} \equiv \text{OR} \equiv \text{if not}$

So $P \text{ unless } Q \equiv P \vee Q. \equiv P \text{ if not } Q.$

NOTE

We cannot apply precedence in English statements, that precedence order was just for mathematical logics only. In english statements, we will have to understand the feelings of the author.

→ Brackets according to feelings.

Ex: (If you study well then you will crack gate) unless you make silly mistakes.

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★ Logical Laws: I am not writing proofs, but can be easily proved with the help of truth table or by case methods.

⇒ Basic Laws:

- Double Negation Rule: The $\neg$ operator cancels itself.

$$\neg(\neg A) = A.$$

- Domination Law:

$$P \vee T = T$$

$$P \wedge F = F$$

$$A \vee \bar{A} = T$$

$$A \wedge \bar{A} = F$$

- Identity Law:

$$P \vee F = P$$

$$P \wedge T = P$$

⇒ Commutative Law: Any operator $\#$ is said to be commutative if $A \# B = B \# A$ for all A, B (Ex: $+$ & $\times$ are comm. in number theory).
Except implication, all operators are commutative.

$$A \wedge B = B \wedge A$$

$$A \vee B = B \vee A$$

$$A \leftrightarrow B = B \leftrightarrow A$$

$$A \oplus B = B \oplus A$$

$$A \uparrow B = B \uparrow A$$

$$A \cup B = B \cup A$$

$$A \rightarrow B \neq B \rightarrow A.$$

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⇒ Associative Law: Operator $\#$ is said to be associative iff $(A \# B) \# C = A \# (B \# C)$ for all A, B, C ($+$, $\times$ are associative in no. theory).

$$(A \vee B) \vee C = A \vee (B \vee C)$$

$$(A \wedge B) \wedge C = A \wedge (B \wedge C)$$

$$(A \leftrightarrow B) \leftrightarrow C = A \leftrightarrow (B \leftrightarrow C)$$

$$(A \oplus B) \oplus C = A \oplus (B \oplus C)$$

So, these 4 operators $\vee$, $\wedge$, $\leftrightarrow$, and $\oplus$ are associative.

$$(A \rightarrow B) \rightarrow C \neq A \rightarrow (B \rightarrow C)$$

$$(A \uparrow B) \uparrow C \neq A \uparrow (B \uparrow C)$$

$$(A \downarrow B) \downarrow C \neq A \downarrow (B \downarrow C)$$

⇒ Idempotence Law: Operator $\#$ is said to be idempotent iff $a \# a = a$ for all a .
(No operator (gm) is idempotent in number theory).

In prop. logic, $\wedge$ & $\vee$ both are idempotent only.

$$A \wedge A = A$$

$$A \vee A = A$$

⇒ Distributive Law: Operator $\#$ is distributive over $*$ iff $a \# (b * c) = (a \# b) * (a \# c)$ for all a, b, c .

(In no. theory $*$ is distributive over $+$, but $+$ is not distributive over $*$).

Here $\wedge$ is distributive over $\vee$ and $\vee$ is distributive over $\wedge$.

So $P \wedge (q \vee r) = (P \wedge q) \vee (P \wedge r)$

$P \vee (q \wedge r) = (P \vee q) \vee (P \vee r)$

Ques: Prove or disprove!

a) $\rightarrow$ is dist. over $\wedge$.

$\Rightarrow$ Let $\alpha = p \rightarrow (q \wedge r)$ and $\beta = (p \rightarrow q) \wedge (p \rightarrow r)$

By case method!

Case I: $p = T$

$\alpha = q \wedge r$

$\beta = q \wedge r$

Case II: $p = F$

$\alpha = T$

$\beta = T$

So $\alpha = \beta$ in each case so Hence Proved.

b) $\wedge$ is distributive over $\rightarrow$.

$\Rightarrow$ Let $\alpha = p \wedge (q \rightarrow r)$ and $\beta = (p \wedge q) \rightarrow (p \wedge r)$

Case I: $p = T$

$\alpha = q \rightarrow r$

$\beta = q \rightarrow r$

Case II: $p = F$

$\alpha = F$

$\beta = T$

$\alpha \neq \beta$ so disproved.

Like this all of the above properties can be proved or disproved.

$\Rightarrow$ De Morgan's Law:

$\neg (p \wedge q) = (\neg p) \vee (\neg q)$

$\neg (p \vee q) = (\neg p) \wedge (\neg q)$

Can also be checked by Examples.

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Ex: P : John is lucky Q : John is cool $P \vee Q$: John is lucky or cool $\neg(P \vee Q)$: John is not lucky and not cool $\equiv (\neg P) \wedge (\neg Q)$ $P \wedge Q$: John is lucky and cool $\neg(P \wedge Q)$: John is not lucky or not cool $\equiv (\neg P) \vee (\neg Q)$

So any expression G containing $(\neg, \vee, \wedge, P, Q, \neg P, \neg Q)$ can be converted to $\bar{G}$ by $(\neg, \wedge, \neg P, \neg Q, P, Q)$.

Ex: $G = (P \wedge \bar{Q}) \vee (\bar{P} \vee Q)$ $\bar{G} = (\bar{P} \vee Q) \wedge (P \wedge \bar{Q})$ $\Rightarrow$ Absorption Law:

$$P \vee (P \wedge Q) = P$$

 $\downarrow$
AndProof: $PT \vee PQ = P$

$$P(T \vee Q) = P$$

$$PT = P$$

$$P = P$$

$$P \vee (\bar{P} \wedge Q) = P \vee Q$$

 $\downarrow$
OrProof: $PT \vee PQ = PT + PQ$

$$P(T \vee Q)$$

 $\Rightarrow$ Implication Law:

$$P \rightarrow Q = \bar{P} \vee Q$$

but $P \rightarrow Q$ is True iff either P is False or Q is True.

⇒ Contrapositive Law:

$$p \rightarrow q = \neg q \rightarrow \neg p$$

★ Simplification Method: Like Truth Table and By case method, this is also one of the method to solve propositional logic questions.

This method use logical laws to solve questions.

Ques: Prove that following statement is a tautology:

a) $\neg p \wedge (p \vee q) \rightarrow q$

⇒ We can do by ^{by} Case method also. But let's do by logical laws.

$$\neg p \wedge (p \vee q) \rightarrow q$$

(Given)

$$(\neg p \wedge p) \vee (\neg p \wedge q) \rightarrow q$$

(Distributive Law)

$$F \vee (\neg p \wedge q) \rightarrow q$$

$$(A \wedge \bar{A} = F)$$

$$\neg p \wedge q \rightarrow q$$

$$(F \vee p = p)$$

Implication Law

$$\neg(\neg p \wedge q) \vee q$$

$$(p \rightarrow q = \neg p \vee q)$$

$$(p \vee \neg q) \vee q$$

(De Morgan's Law)

$$p \vee \neg q \vee q$$

$$p \vee T$$

$$(q \vee \bar{q} = T)$$

$$T$$

$$(p \vee T = T)$$

So, it is a tautology.

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$$b) (p \wedge q) \rightarrow (p \vee r)$$

$$\Rightarrow (p \wedge q) \vee (p \vee r)$$

(Implication Law)

$$\bar{p} \vee \bar{q} \vee p \vee r$$

(De Morgan's Law)

$$T \vee \bar{q} \vee r$$

(TV anything = T)

$$\underline{\underline{T}}$$

So Tautology

$$c) p \rightarrow (r \rightarrow p)$$

(Not writing Names)

$$p \rightarrow (\bar{r} \vee p)$$

$$\bar{p} \vee \bar{r} \vee p$$

$$T \vee \bar{r}$$

$$\underline{\underline{T}}$$

$$d) \bar{r} \vee (\bar{r} \rightarrow p)$$

$$\bar{r} \vee r \vee p$$

$$T \vee p$$

$$\underline{\underline{T}}$$

Que: Let p & q be statements. Determine each of the following in tautology, contradiction, or neither.

$$a) p \leftrightarrow \neg(\neg p)$$

$$\Rightarrow p \leftrightarrow p$$

$$= \underline{\underline{T}}$$

(Tautology)

$$b) p \wedge \neg p$$

$$= \underline{\underline{F}}$$

(Contradiction)

$$c) p \vee \neg p$$

$$= \underline{\underline{T}}$$

(Tautology)

$$d) (p \wedge q) \vee (\neg p \wedge \neg q)$$

$$\Rightarrow [p \vee (\neg p \wedge \neg q)] \wedge [q \vee (\neg p \wedge \neg q)]$$

$$\Rightarrow (p \vee \neg p) \wedge (p \vee \neg q) \wedge (q \vee \neg p) \wedge (q \vee \neg q)$$

$$T \wedge (p \vee \neg q) \wedge (q \vee \neg p)$$

(can solve by case method)

$$\text{Case 1: } p = T$$

$$\underline{q} \rightarrow \underline{\text{So Contingency.}}$$

* Analysis of Implication:

Ques: " $\alpha \rightarrow \beta$ " is tautology or not?

$\Rightarrow$ if $\alpha = \text{True}$ and $\beta = \text{False}$, can be proved by any method then it is not tautology. And if proving this is not possible then it is tautology.

So when $\alpha = T$ and $\beta = F$ is never possible then it is tautology. Now there are two approaches:

Approach 1

make $\alpha = \text{True}$ and then try your best to make $\beta = \text{False}$.

If possible then Not Tautology else tautology.

Approach 2

make $\beta = \text{False}$ and then try your best to make $\alpha = \text{True}$.

If possible, then not tautology else tautology.

Ex: 1 $(p \wedge q) \rightarrow (p \vee q)$ is tautology?

$\Rightarrow$ let $\alpha = p \wedge q$ and $\beta = p \vee q$.

make $\alpha = \text{True} \Rightarrow$ only possible if $p = T, q = T$

and $\beta = \text{True}$ in this case. So not possible to

make β false by $p = T$ & $q = T$. So Tautology.

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Ex:2 $(p \vee q) \rightarrow (p \oplus q)$

α β

$$\alpha = T \Rightarrow p = T, q = T$$

$$\text{then } p = F$$

So possible so not Tautology (Used Approach 1)

Ques: Check if following expression is Tautology or not?

$$[(p \rightarrow (q \rightarrow R)) \wedge (p \vee q)] \rightarrow R$$

→ Method One (Truth Table)

P	q	R	$[(p \rightarrow (q \rightarrow R)) \wedge (p \vee q)] \rightarrow R$
F	F	F	T
F	F	T	T
F	T	F	T
F	T	T	T
T	F	F	T
T	F	T	T
T	T	F	T
T	T	T	T

So, Tautology

→ Method 2 (BS Case Method)

Case I: $R = T$

$$[(p \rightarrow (q \rightarrow R)) \wedge (p \vee q)] \rightarrow \text{True}$$

$$(p \rightarrow (q \rightarrow T)) \wedge (p \vee q) \rightarrow T$$

$$(p \rightarrow T) \wedge (p \vee q) \rightarrow T$$

$$T \wedge (p \vee q) \rightarrow T$$

$$p \vee q \rightarrow T$$

Case II: $R = F$

$$\text{Anything } \rightarrow F = T$$

So ~~False~~ True

Method 2: By Case Method.

Case I $x = \text{True}$

Anything $\rightarrow \text{True} = \text{True}$

So True.

Case II $x = \text{false}$

Anything $\rightarrow \text{false}$

$(p \rightarrow (q \rightarrow \text{false})) \wedge (p \wedge q) \rightarrow \text{false}$

$(p \rightarrow \bar{q}) \wedge (p \wedge q) \rightarrow (q \rightarrow \text{false} = \bar{q})$

$(\bar{q} \vee \bar{p}) \wedge (p \wedge q) \rightarrow \text{f}$

$(\bar{q} \vee \bar{p}) \wedge (p \wedge q)$

$(\bar{q} \vee \bar{p}) \vee (\bar{p} \wedge q)$

$(\bar{q} \wedge \bar{p}) \vee (\bar{p} \vee \bar{q})$

Case 2.1, $q = \text{True}$

$p \vee \bar{p}$

$= \text{True}$

Case 2.2, $q = \text{False}$

$\bar{p} \vee \bar{p} \vee \text{True}$

$= \bar{p} \vee \text{True}$

So Tautology.

Method 3: By logical law

Method 4: By implication analysis.

$\alpha = [(p \rightarrow (q \rightarrow R)) \wedge (p \wedge q)]$

$\beta = R$

by App 2: $R = \text{False}$ ($\beta = \text{False}$)

Now prove $\alpha = \text{True}$ or else it will be Tautology.

$(p \rightarrow (q \rightarrow \text{False})) \wedge (p \wedge q)$

$(p \rightarrow \bar{q}) \wedge (p \wedge q)$

$(\bar{q} \vee \bar{p}) \wedge (p \wedge q)$

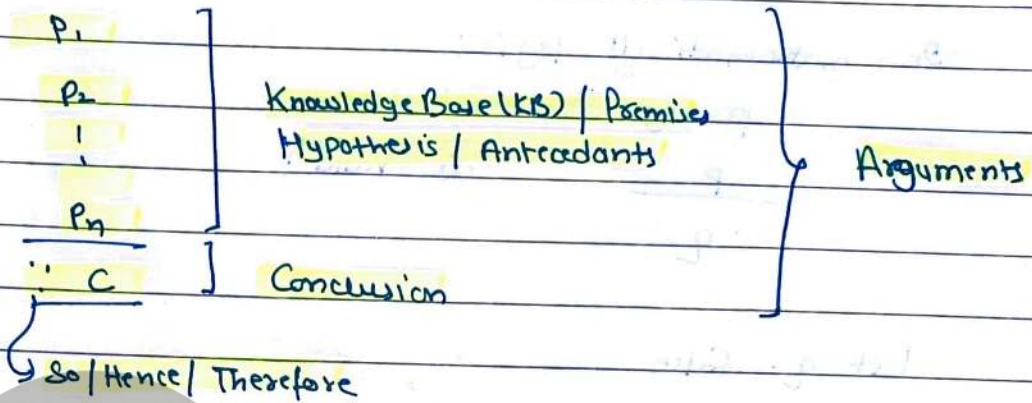
$(\bar{q} \vee \bar{p}) \wedge (\bar{p} \vee \bar{q})$

False

$(A \wedge \bar{A} = \text{False})$

So Tautology.

* Logical Argument: It is just a sequence of statements which contains knowledge base (set of premises) and a conclusion.



⇒ Validity of Argument: An argument is valid iff we assume all premises to be true, then conclusion must be True. In other words, Argument is invalid iff it is possible that all premises are true, still conclusion is false.

So, Argument is valid iff

$$A \wedge B \wedge C \rightarrow D \quad \text{where } A, B, C \text{ are KB \& } D \text{ is conc.}$$

↘ is tautology

Invalid if this is not a tautology.

⇒ Method to check if given argument is valid or not?

• Make the conclusion false, now try to make All premises True.

• If possible, then Argument is Invalid.

• Else argument is Valid.

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Ex:1 If it is raining, He will take umbrella.

It is raining.

 $\therefore$ He will take umbrella.

In mathematical logic:

$$p \rightarrow q$$

$$\frac{p}{\therefore q}$$

is valid?

Let $q = \text{False}$ $p = \text{True}$ $p \rightarrow q = \text{False}$

, Not possible to make all premises True by making conclusion false.

So, Valid Argument.

Also intuitively if p then q and then we have p then obviously q is True.So, $p \rightarrow q$

$$\frac{p}{\therefore q}$$

is valid argument form.

(Very famous called modus ponens)Ex:2

$$p \vee q$$

$$\frac{p}{\therefore q}$$

valid?

No $\Rightarrow$ Intuitively p or q and we have p then q not guaranteed to be true so not valid.

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let $q = F$
 $p = T$
 $p \vee q = \text{True}$

As all premises are T even if conclusion is False.
 So, this is Invalid form of argument.

Ex:3
$$\begin{array}{l} p \vee q \\ \underline{\neg p} \\ \hline \therefore q \end{array}$$
 This is Valid.

Ex:4
$$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$$
 Valid

★ Rules of Inference: Some popular, frequently used, Valid argument. (They also have a name).
 (Names are not important) Some are: $\neg$ (All are valid)

• Modus Ponens:
$$\begin{array}{l} p \rightarrow q \\ \underline{p} \\ \hline \therefore q \end{array}$$

• Modus Tollens:
$$\begin{array}{l} p \rightarrow q \\ \underline{\neg q} \\ \hline \therefore \neg p \end{array}$$

• Conjunctive Simplification:
$$\begin{array}{l} p \\ q \\ \hline \therefore p \end{array}$$

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• Disjunctive Syllogism:

$$\begin{array}{l} p \vee q \\ \underline{\neg q} \\ \therefore p \end{array}$$

• Hypothetical Syllogism:

$$\begin{array}{l} p \rightarrow q \\ \underline{q \rightarrow r} \\ \therefore p \rightarrow r \end{array}$$

• Addition

$$\begin{array}{l} p \\ \therefore p \vee q \end{array}$$

• Conj. simplification (Alternate)

$$\begin{array}{l} p \wedge q \\ \therefore p \end{array}$$

• Resolution:

$$\begin{array}{l} p \vee q \\ \underline{\neg p \vee r} \\ \therefore q \vee r \end{array}$$

(All these can be understood by intuition. No need to byheart. Also formal proof is good).

★ The inference symbol ($\models$ or $\vdash$):

$KB \models \gamma$ means. KB infers γ .

i.e. $\frac{KB}{\therefore \gamma}$ } logically Valid Arg.

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So, if $\left. \begin{array}{c} P_1 \\ P_2 \\ \vdots \\ P_n \\ \hline C \end{array} \right\}$ valid Arg. , then $P_1, P_2, P_3, \dots, P_n \vdash C$
 , And $P_1 \wedge P_2 \wedge P_3 \wedge \dots \wedge P_n \rightarrow C$ is a tautology

So $\nVdash$ means invalid argument and does not infer.

$KB \models Y$ can be read as:

- KB infers Y .
- KB entails Y .
- KB implies Y .
- Y is a consequence of KB .

Ques: If $KB = \{p \rightarrow q, p, s \rightarrow r\}$, then which of the following are True?

☒ (a) $KB \models p$

☒ (b) $KB \models q$

☒ (c) $KB \models s$

☒ (d) $KB \models r$

$\Rightarrow$ Can check by formal method or by intuition.

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For any questions or issues regarding my handwritten notes, feel free to reach out via email at karan757527@gmail.com or DM me on telegram at @karan757527 (or by clicking [here](#)).

I would love to know if you liked my notes. Your feedback and appreciation are invaluable, so don't hesitate to share your thoughts. Click [here](#) to connect with me on LinkedIn or you can find me as @agrawalkaran.

Wishing you success on your journey ahead!

- Karan Agrawal